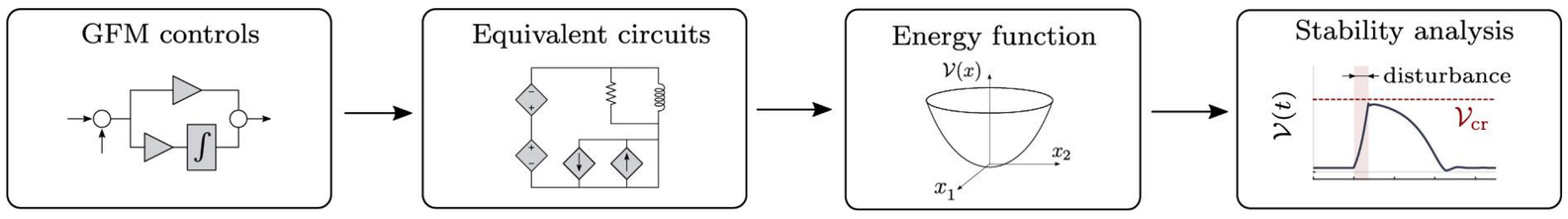
Large-signal Stability Assessment of Grid-forming (GFM) Inverters with Equivalent-circuit Models

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Motivation

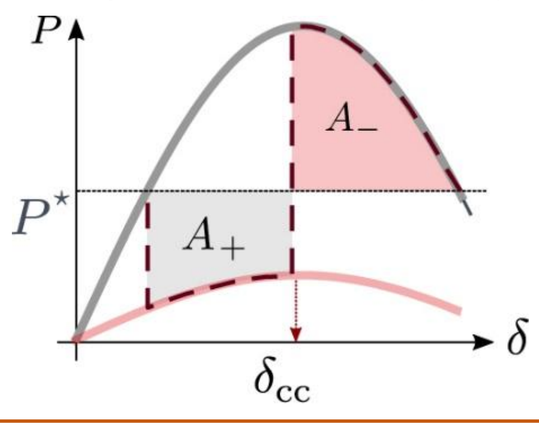
- Equal area criterion and Lyapunov energy function-based methods are traditionally used for machine-based systems
- Can we use classical tools for large-signal stability assessment of GFM inverters?

Our proposed approach for GFM inverters

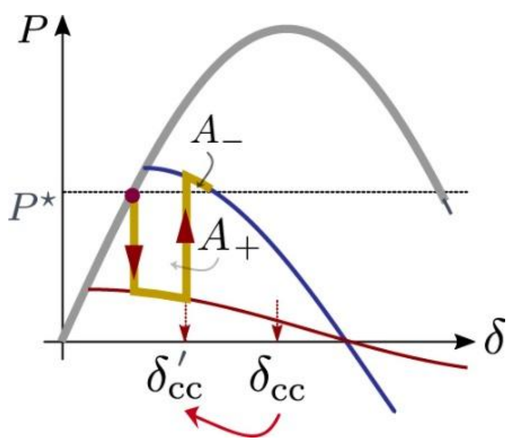


Limitations of classical stability assessment methods

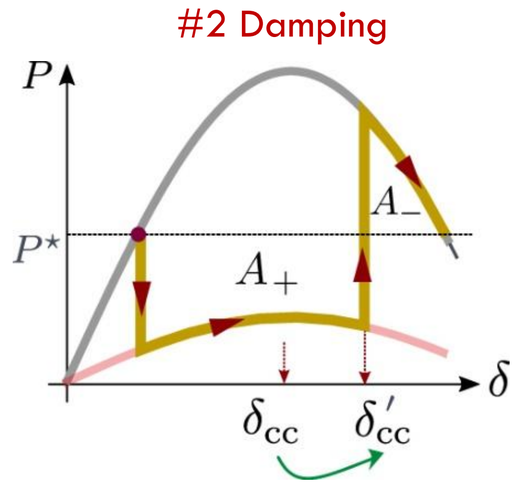
Equal area criterion (EAC)



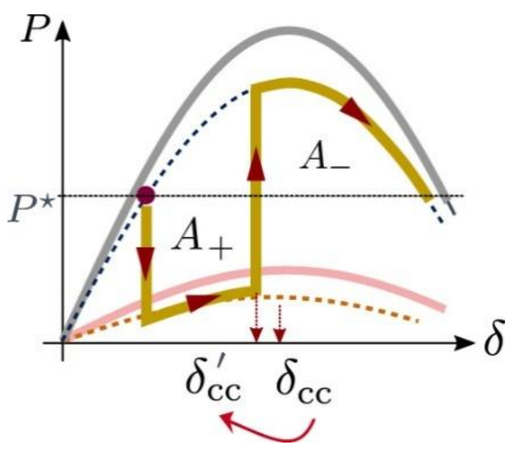
#1 Current limiters



#2 Damping



#3 Q-V droop



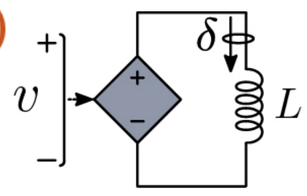
Development of the equivalent circuit for the primary GFM controller

1 For droop controller,

$$\omega = \omega_o - m_p \frac{\omega_c}{s+\omega_c}(p - p^\star)$$

$$\frac{1}{m_p \omega_c}\frac{d^2\delta}{dt^2} + \frac{1}{m_p}\frac{d\delta}{dt} = p^\star - p$$

2



if $i = \delta$ and $L = 1$

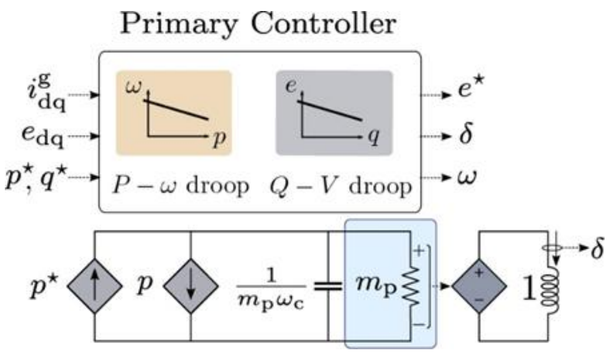
$$v = L\frac{d\delta}{dt} \qquad \frac{d^2\delta}{dt^2} = \frac{dv}{dt}$$

3

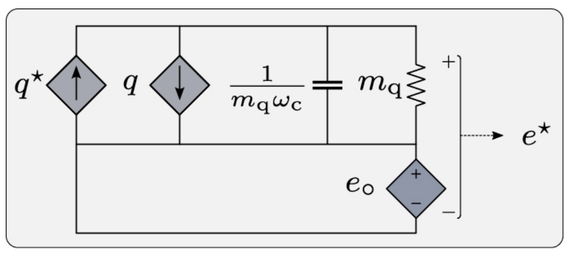
$$\underbrace{p^\star - p}_{i_s} = \underbrace{(m_p\omega_c)^{-1}}_{C_{eqv}}\frac{dv}{dt} + \underbrace{(m_p)^{-1}}_{(R_{eqv})^{-1}}v$$

$$M = C_{eqv} = (m_p\omega_c)^{-1} \qquad D = R_{eqv}^{-1} = m_p^{-1}$$

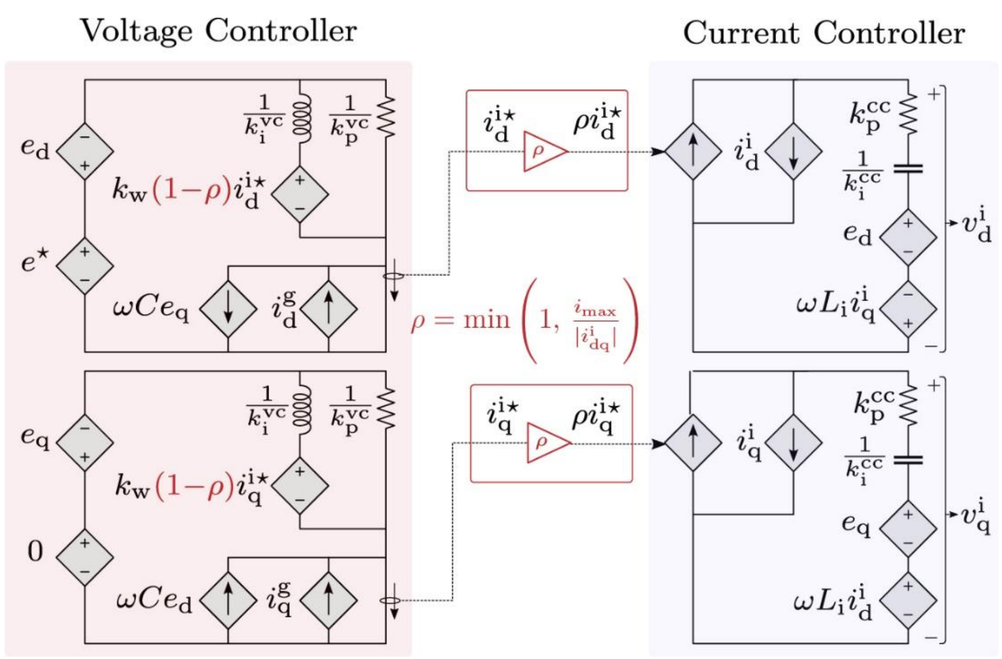
A comprehensive equivalent circuit for the cascaded GFM controller



Solution for limitation #1: Damping

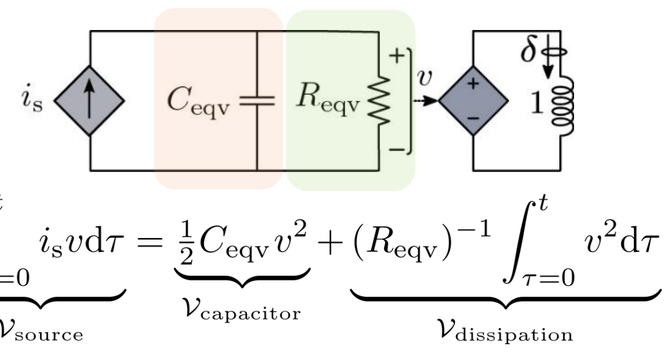


Solution for limitation #2: Q-V droop

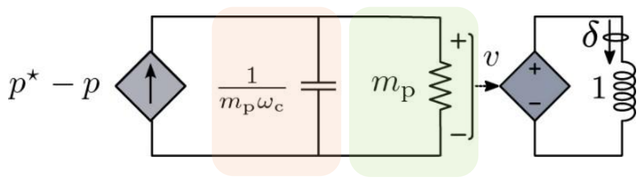


Solution for limitation #3: Current limiting

Development of the energy function

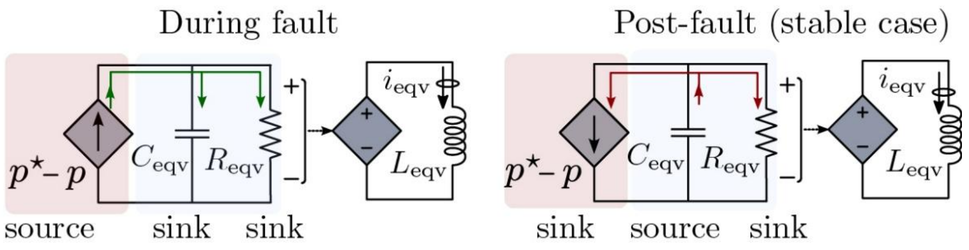


$$\underbrace{\int_{\tau=0}^{t} i_s v\,d\tau}_{\mathcal{V}_{source}} = \underbrace{\tfrac{1}{2}C_{eqv}v^2}_{\mathcal{V}_{capacitor}} + \underbrace{(R_{eqv})^{-1}\int_{\tau=0}^{t} v^2 d\tau}_{\mathcal{V}_{dissipation}}$$

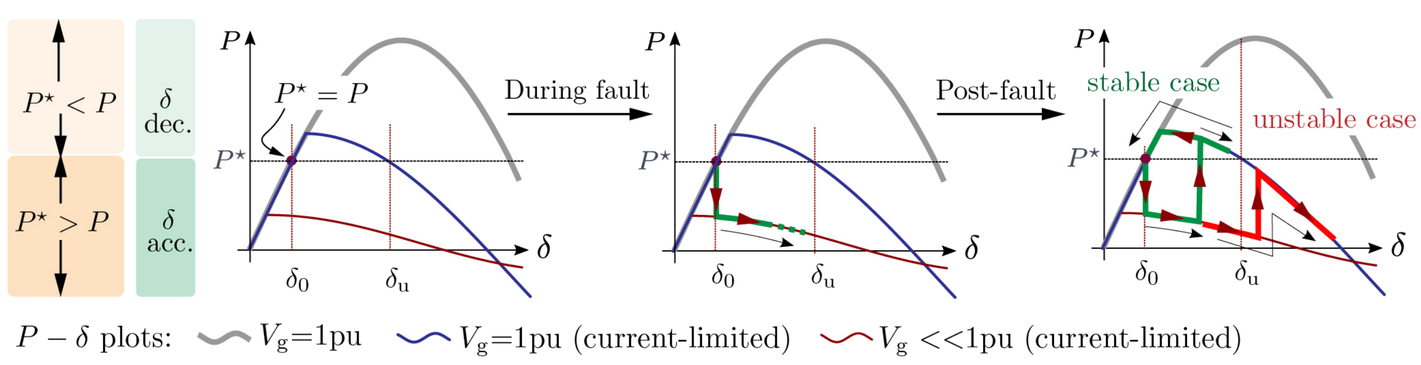


$$\int_{\tau=0}^{t} \underbrace{(p^\star - p)}_{i_s} v\,d\tau = \tfrac{1}{2}\underbrace{(m_p\omega_c)^{-1}}_{C_{eqv}}v^2 + \underbrace{m_p^{-1}}_{(R_{eqv})^{-1}}\int_{\tau=0}^{t} v^2 d\tau$$

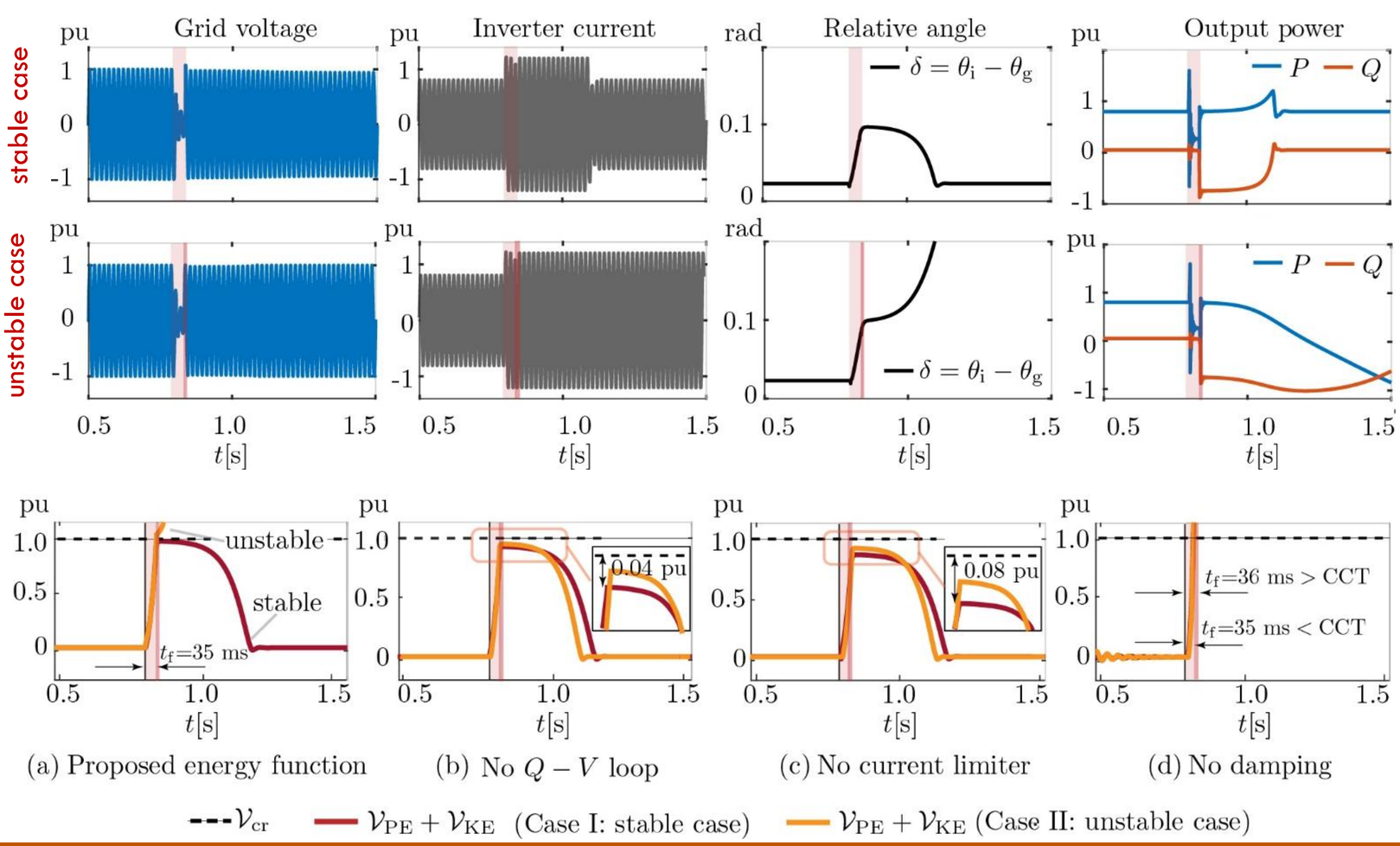
Power-angle dynamics through the lens of the proposed equivalent circuit



The trajectory of the power-angle curve of a GFM inverter is explained using the flow of current in the proposed equivalent circuit



Validation through simulation results (undervoltage fault)



Conclusion

- This work provides **physical insight into the mechanisms of energy exchange in GFM inverters** and establishes their connection to transient stability
- This modeling and analysis **paradigm seamlessly overcome limitations of classical approaches**

References

[1] N. Baeckeland, D. Pal, G.-S. Seo, B. Johnson, and S. Dhople, "Large-signal stability analysis of grid-forming inverters with equivalent-circuit models," *IEEE TEC 2025*

[2] D. Chatterjee, N. Baeckeland, G.-S. Seo, B. Johnson, and S. Dhople, "Demystifying Grid-forming Inverter Large-signal Stability Assessment with Equivalent-circuit Models," *IEEE COMPEL 2025*

Acknowledgements

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